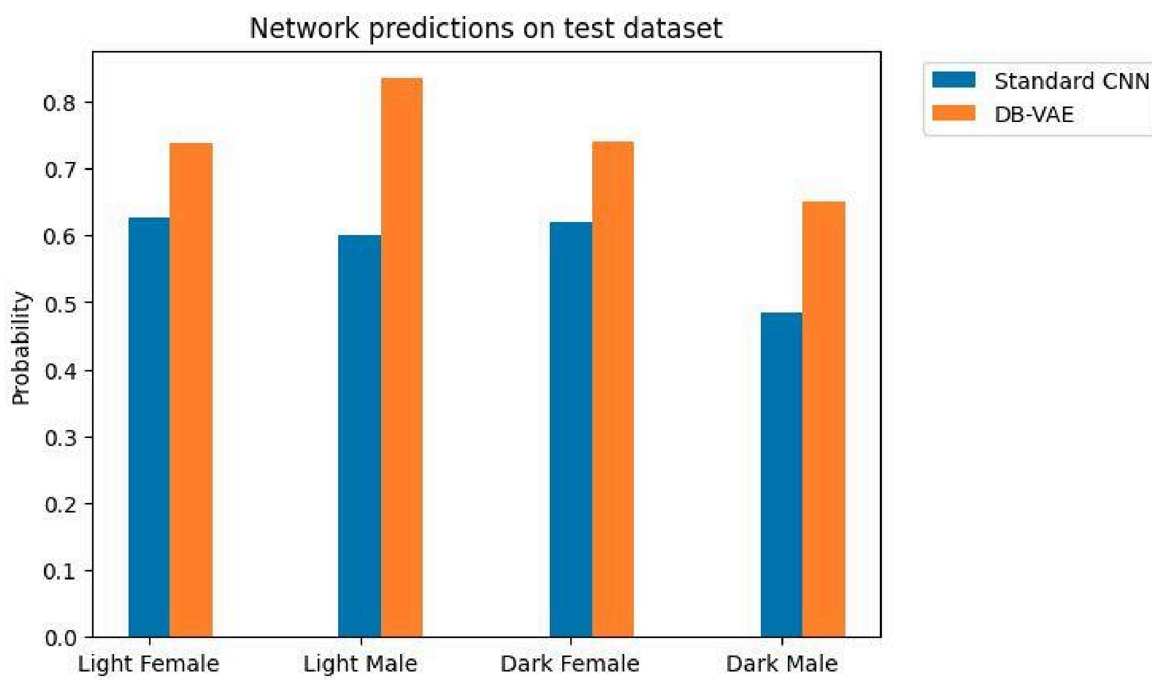
- How does the accuracy of the DB-VAE across the four demographics compare to that of the standard CNN? Do you find this result surprising in any way?
    - DB-VAE accuracy is increased across all four demographics. While I expected this for lower accuracy demographics (black male) in standard CNN, an increase in higher accuracy (white female) demographics with DB-VAE did surprise me. I think it is because of overfitting of data for that demographic in standard CNN. If the number of epochs are increased, same overfitting results in decreased accuracy in the DB-VAE as well.

- Can the performance of the DB-VAE classifier be improved even further?
    - I was able to increase the accuracy of the DB-VAE by increasing the smoothing_fac to 0.01 from 0.001




- In which applications (either related to facial detection or not!) would debiasing in this way be desired? Are there applications where you may not want to debias your model?
    - Yes I think it is required to debias the models unless the application is nuanced. For example, a lot of bias is towards English language models, whereas other languages are largely ignored either in interest of volume of data or number of users able to understand English.

- Do you think it should be necessary for companies to demonstrate that their models, particularly in the context of tasks like facial detection, are not biased? If so, do you have thoughts on how this could be standardized and implemented?
    - Yes, it is essential for companies to debias their models since it is increasing accuracy of detection and recognition as well. This could be standardized by having a standard test set that all companies have to pass a minimum score on. Like the Turing test for AI.

- Do you have ideas for other ways to address issues of bias, particularly in terms of the training data?
    - Correction using GANs is another way to address bias. The difference in predicted and observed values can be decreased and fed back into the encoder.